Fluid Flow Through Fluidized Beds

Pre-Lab Plan
Unit Operations Lab 2
2/4/2026

Group #1, Tuesday 1PM section

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1. Experimental Objective

The objective of the laboratory experiment is to investigate the fluidization behavior of silica sand and ion exchange resin beds by recording their pressure drops as a function of gas velocity under increasing/decreasing flow conditions. The experiment involves packed and fluidized bed theories, using both the Ergun and force-balance equations to evaluate the measured pressure data as compared to the literature.

2. Experimental

2.1 Apparatus

Image Sourced from lab describing procedure in Ref 5.

1. **Fluidized Bed Vessel**- The 4” plexiglass tube on the left is the active column used for the experiment

2. **Porous Disc (bed Support)**- The metal section at the bottom of the plexiglass tube holding sand/resin whilst allowing the air to pass through.
3. **Entrainment Disengagement Zone**- The cone shaped section on top of the plexiglass tube is to reduce the loss of particles the air carries out,
4. **Air Rotameters**- The two vertical glass tubes on the far left and far right sides are to measure the air flow rate in the system.
5. **Air Control Valves**- The red hand-wheels near the base of the pipe used to adjust the air flow going into the columns
6. **Pressure Gauge**-The dial seen near the left-hand piping to observe the air pressure upstream.
7. **Manometer**-The long scale between the two columns used to measure the pressure drop across the bed.
8. **Sieves**- They are on the shelf below the board; they are used for the first part of the particle size distribution analysis.

2.2 Materials and Supplies

Material and Supplies	Use within Experiment	Material/Supply
Sand	Sand will be used in the fluidized bed experiment.	Supply
Resin	Resin will be used in the fluidized bed experiment.	Supply
Sifts	Separates the different sizes of resin and sand particles to measure the average grain size.	Material
Air flow	Used to conduct the experiment and see the particle effect in passing air.	Supply
Containers	Used to hold sand and resin for the experiment,	Material
Scales	Measures amount for sand and resin used in the experiment. Also used in particle size calculations.	Material
Spatulas	Used to acquire the amount desired of sand or resin from storage containers.	Material
Vacuum	Used for cleaning the floor and environment from dust released from the experiment.	Material

2.3 Experimental Plan

<u>PROCEDURE</u>:

1. First perform measurements to determine the porosity of both bed materials
   a. To do this, measure out 100 mL of material in a graduated cylinder
   b. Next, add water slowly to the sand until water reaches the surface of the material. Be sure to not leave any air gaps by tapping on the cylinder gently. Keep track of the total amount of water added.
   c. Divide the total amount of water added by the material's starting volume to get the porosity.
2. Once the porosity is calculated for both materials, choose one material to start with and charge the vessel. Obtain a height of 6".
3. Slowly increase the air flowrate and record the pressure drop and bed height once the material's level is static. Do this for every increase.
4. Continue this process until there is an insignificant pressure drop seen in the manometer. This indicates that the minimum fluidization level has been surpassed. This is the goal.
5. Repeat this process for the second material. Also, re-do the experiment for both materials, but this time, start with a height of 10".

<u>Independent Variables</u>: physical properties of materials, air flowrate, material porosity, viscosity & density of air, initial bed height, temperature, etc.

<u>Dependent Variables</u>: pressure drop, bed height throughout experiment, minimum fluidization velocity, fluidization behavior, etc.

<u>POST-LAB ANALYSIS</u>:

1. Create a graph showing the relationship of pressure drop over length ($\frac{\Delta P}{L}$) over gas flow rate for all resin and sand trials.
2. Utilizing graph trends determines the minimum fluidization velocity from the graph created for resin and sand.
3. Calculate the theoretical minimum fluid velocity utilizing equation 7.51 from (McCabe, Smith, and Harriott, Unit Operations of Chemical Engineering, 7th ed. Pg 167).
4. Evaluate and compare theoretical values of minimum fluid velocity for both sand and resin.

5. Prove that the Ergun equation ($\frac{\Delta P}{L} = \frac{150 \overline{V}_0 \mu}{\phi_s^2 D_P^2} \frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{1.75 \rho \overline{V^2}_0}{\phi_s^{D_P^{\square}}} \frac{1-\varepsilon}{\varepsilon^3}$) with data points to predict the value of pressure drop,
6. Compare and access if data is accurate to experimental data analysis from the plots.

Project Safety Evaluation

Project Title: Fluid Flow Through Fluidized Beds

Date: 2/4/26

	Yes/No	
Potential electrical hazards?	No	If yes, identify conditions:
Potential mechanical hazards?	No	If yes, identify conditions:
Potential pressure hazards?	Yes	If yes, identify conditions: **Air used to measure the system can exhibit pressures of up to psia.**
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards are involved: **Resin can cause irritation when exposed to skin and eyes.**
Gloves required?	Yes	If yes, specify type: **Nitrile gloves must be used when handling resin and gloves.**
Other hazards	Yes	If yes, specify conditions: **Goggles must be used for dust that can fly in the air. Masks are optional as it depends on the person conducting the experiment.**
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: **Resin CAS: 25085-99-8 "resin"**
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	N/A	
Pressure (psia)	128 psia	Pressurized air traveling into the bottom of the bed.
Safety Controls	*yes/no*	*If yes, Provide description*
Regulator	Yes	Keeps the air supply at a safe and effective pressure.

REFRENCES:

1. W. McCabe, J. Smith, and P. Harriot, Unit Operations of Chemical Engineering, 7 th ed. McGraw-Hill, 2005.
2. J. Davidson, R. Clift, and D. Harrison, Eds., Fluidization, 2 nd ed. Academic Press, 1995
3. D. Kunii and O. Levenspiel, Fluidization Engineering, 2 nd ed. Elsevier Science and Technology, 1992.
4. R. H. Perry and D. W. Green, Eds., Perry's Chemical Engineer's Handbook, 7 th ed. McGraw-Hill, 1997.
5. ChE 382 Unit Operations Laboratory Manual, Fluid Flow Through Fluidized Beds, rev. Spring 2023v2.

Safety Statement

I have read relevant background material for the Unit Operations Laboratory entitled: "Fluid Flow Through Fluidized Beds" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Team Member Signature: Johsua Laforest — Date: 4 Feb 2026

Team Member Signature: Ify Bidokwu — Date: 4 Feb 2026

Team Member Signature: Cesearo Dilosa — Date: 4 Feb 2026

Team Member Signature: Javier Silva — Date: 4 Feb 2026